

Eric Balch

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EDUCATION

University of California, Los Angeles

Sept. 2024 – June 2028

Bachelor of Science in Mechanical Engineering

GPA: 3.95/4.0

- Extracurriculars: Bruin Formula SAE Racing, Bionics Lab, Tau Beta Pi

EXPERIENCE

Mechanical Engineering Intern

June 2026 – Present

ANELLO Photonics Inc.

Santa Clara, CA

- Electromechanical design, development, assembly, & testing of Inertial Navigation System calibration fixtures
- Selection & integration of high-precision motors and encoders for repeatable actuation
- Develop 2D part & assembly drawings w/ precise GD&T callouts for manufacturing and internal documentation
- Design breakout cable tester (KiCad) to validate harness functionality & streamline manufacturing quality control
- Increased unit calibration throughput by 250% while maintaining mG level performance accuracy

Drivetrain Lead & Responsible Engineer

Oct. 2024 – Present

Bruin Formula SAE Racing Drivetrain Team

Los Angeles, CA

- Directing drivetrain architecture, leading team of 10 through full design cycle
- Design (SolidWorks) front/rear sprockets and motor shaft w/ Design-for-Manufacture (DFM) emphasis
- Iteratively FEA-test (ANSYS Workbench) part geometry to optimize slotting & reduce weight by >10% while withstanding extreme mechanical stress
- Develop GD&T drawings & communicate specs to local sponsors for outsourcing component manufacturing
- Create and execute (F360 CAM) CNC toolpaths for manufacturing of motor shaft and motor mounts, reducing manufacturing costs by >25%
- Reduced manufacturing time for differential and motor mounts by >20% by transitioning from soft jaws to tabbing
- Designed (SolidWorks) 80/20 assembly to stress test motor-to-sprocket chain (Up to 4500 N)

Undergraduate Researcher

Nov. 2025 – Present

UCLA Bionics Laboratory

Los Angeles, CA

- Design (SolidWorks) endoscope with independently controlled camera and surgical accessories to improve polyp resection and ease of use for surgeons
- Focus on the distal component of the device, creating the housing and operational mechanism for the accessory attachment point
- Rapid prototyping of designs using 3D printed components & off-the-shelf tubing & control wires

Biomechanical Engineering Intern

June 2023 – Sept. 2023

Stanford University

Stanford, CA

- Developed computational fluid dynamics (CFD) simulation models of diseased coronary arteries
- Ran CFD simulations evaluating grafts with complex BCs to consider the effects of multiple stenoses
- Modeled pipeline for cardiovascular anatomic models of patient-specific coronary bypass procedures to analyze changing hemodynamics of different attachment locations

Lifeguard and Swim Instructor

June 2021 – Sept. 2025

City of Mountain View

Mountain View, CA

- Head Guard leadership role supervising a staff of up to 5 lifeguards, CPR & water rescue certified

RELEVANT PROJECTS

Team Lead & Lead Designer, Engineering Lab Course

Sept. 2022 – June 2024

- Designed and built winning wireless car parking lot sensors & display for CTE Capstone project
- Designed and built winning robots for two Vex Robotics timed competitive challenges
- Designed and built hydraulic arm to perform timed competitive precision challenges
- Coordinated team member contributions and presented product design results to class

TECHNICAL SKILLS

Software: SolidWorks, Fusion 360, ANSYS, MS Office, Gemini, MATLAB, Python, Java, KiCad

Tools: Lathe & Mill (CNC & Manual), Drill Press, Band Saw, 3D Printer, Laser Cutter, Soldering